

Post Graduate Common Entrance Test - 2025

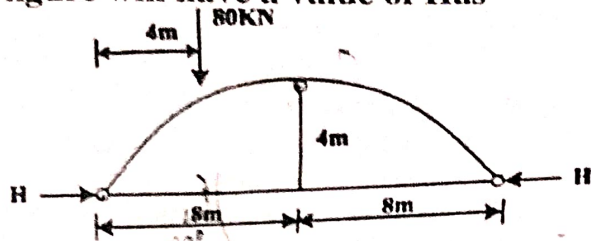
CIVIL ENGINEERING

Time : 2 hrs

Version Code :- D1

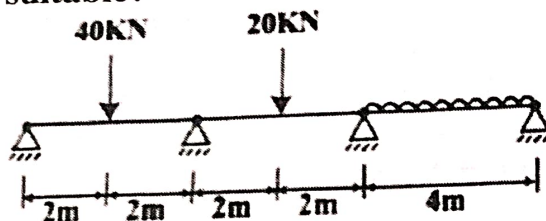
Max Marks : 100

1. The three hinged arch shown in the figure will have a value of Has



- (1) 20 kN (2) 30 kN
(3) 40 kN (4) 50 kN

2. For analysing the continuous beam shown in the figure, which of the following method will be most suitable?



- (1) Force method
(2) Displacement method
(3) Both (1) and (2)
(4) None of the above

3. The maximum area of tension reinforcement in beam shall not exceed

- (1) 0.04 bD
(2) 0.02 bD
(3) 0.08 bD
(4) 0.10 bD

4. For RCC column that is effectively held in position and restrained against rotation at one end and at the other end restrained against rotation but not held in position, the effective length is

- (1) 1.20
(2) 0.80
(3) 1.00
(4) 1.50

5. In a pedestal, the factor by which the effective length should not exceed the least lateral dimension is

- (1) 02
(2) 03
(3) 04
(4) 05

6. For the purpose of design as per IS-456, deflection of RC slab or beam is limited to

- (1) 0.2% of span
(2) 0.25% of span
(3) 0.4% of span
(4) 0.45% of span

7. Which of the following is generally not designed for shear?

- (1) A slab
(2) A cantilever beam
(3) A footing
(4) None of the above

8. The addition of coarser particled soil like sand or silt to clay, causes
- (1) Decrease in liquid limit as well as in plasticity index
 - (2) Decrease in liquid limit and increase in plasticity index
 - (3) Increase in liquid limit as well as in plasticity index
 - (4) Decrease in liquid limit and no change in plasticity index
9. Shrinkage limit of clay whose void ratio in dry state is 0.55, specific gravity is 2.75 is
- (1) 5%
 - (2) 10%
 - (3) 20%
 - (4) 15%
10. In a drained triaxial compression test, a saturated specimen of a cohesionless sand fails under a deviator stress of 3 kg/cm^2 when the cell pressure is 1 kg/cm^2 . The effective angle of shearing resistance of sand is about
- (1) 64
 - (2) 45
 - (3) 37
 - (4) 18
11. The net ultimate bearing capacity of a purely cohesive soil is
- (1) depends on width of footing and is independent of depth of footing
 - (2) depends on width of footing and is independent of width of footing
 - (3) depends on both depth and width of footing
 - (4) is independent of both depth and width of footing
12. For a pure cohesive soil, the bulk unit weight is 20 KN/m^3 and a cohesion of 12.5 KPa . A vertical cut in this soil can temporarily stand without any lateral support for a height of
- (1) 1.25 m
 - (2) 2.5 m
 - (3) 3.2 m
 - (4) 5 m
13. Bulb of pressure denotes
- (1) Contour of equal vertical stress
 - (2) Contour of equal shear stress
 - (3) Contour of equal radial stress
 - (4) None of the above
14. A normally consolidated clay is settled by 2 cm when effective stress was increased from 100 KPa to 200 KPa . Calculate the settlement when the effective stress is increased to 400 KPa .
- (1) 2 cm
 - (2) 4 cm
 - (3) 6 cm
 - (4) 4.5 cm
15. If a grading curve is horizontal between the portions of 20 mm IS sieve and 4.75 mm IS sieve, the graded aggregates do not contain
- (1) 20 mm particles
 - (2) 10 mm particles
 - (3) 4.75 mm particles
 - (4) All of the above
16. Total solids present in water comprise of total dissolved solids and
- (1) Settleable solids
 - (2) Suspended solids
 - (3) Fixed solids
 - (4) Volatile solids

17. The rate of filtration of slow sand filters ranges from
(1) 100-200 lt/hour/sq.m
(2) 40-50 lt/hour/sq.m
(3) 250-350 lt/hour/sq.m
(4) None of the above
18. The colour of water for domestic supplies on standard platinum-cobalt scale should not exceed
(1) 10 units
(2) 15 units
(3) 25 units
(4) 20 units
19. The dissolved oxygen content of waste water may be determined by
(1) Winkler's method
(2) Oxidation method
(3) Reflux method
(4) None of the above
20. The nitrates are reduced to nitrogen gas either by autotrophic or heterotrophic anaerobic bacteria. This process is called as
(1) De-nitrification
(2) Nitrification
(3) Oxidation
(4) None of the above
21. The procedure that is used to increase the solids content of sludge by removing a portion of liquid fraction is termed as
(1) Thickening of sludge
(2) Heat drying of sludge
(3) Conditioning of sludge
(4) None of the above
22. The design period for the design of a water supply project is generally taken as
(1) Less than 10 years
(2) 20-30 years
(3) 50 years
(4) More than 50 years
23. The hydraulic loading rate for low rate filters are
(1) 2-5 m³/d/m²
(2) 8-10 m³/d/m²
(3) 1-4 m³/d/m²
(4) None of the above
24. Friction components that are considered for the calculation of super elevation is
(1) Longitudinal friction
(2) Lateral friction
(3) Brake efficiency
(4) Braking distance
25. "GRAVITY MODEL" is used in transportation planning process for
(1) Model split
(2) Trip Distribution
(3) Trip Generation
(4) Trip Assignment
26. Benkelman Beam Deflection method is used for the design of
(1) Rigid overlay on rigid pavement
(2) Flexible overlay on flexible pavement
(3) Flexible overlay on rigid pavement
(4) Rigid overlay on flexible pavement

27. Conflict which may occur in a Rotary intersection is
(1) Merging and Diverging
(2) Crossing and Merging
(3) Crossing and Diverging
(4) Crossing, Merging and Diverging
28. The main objective of providing ballast in the railway track is
(1) Uniform distribution of load from sleepers over large area
(2) To hold sleepers in position
(3) To provide elasticity and resilience to the track
(4) All of the above
29. Rail chairs are used to fix
(1) Flat footed rails
(2) Bull headed rails
(3) Double headed rails
(4) None of the above
30. What is the psychological widening of a pavement on a horizontal curve of radius 230 m for a design speed of 80 kmph?
(1) 0.455 m
(2) 0.555 m
(3) 0.655 m
(4) 0.755 m
31. In a Bituminous pavement, alligator cracking is mainly due to
(1) Inadequate wearing course
(2) Inadequate thickness of subbase course of pavement
(3) Use of excessive bituminous material
(4) Fatigue arising from repeated stress applications
32. Hydrology is the science which deals with water
(1) in atmosphere
(2) on the ground
(3) on earth surface
(4) on earth's surface and beneath it
33. Precipitation is same as
(1) rainfall
(2) snowfall
(3) hail
(4) all of the above
34. Thiessen polygon is used in connection with
(1) average rainfall determination
(2) catchment area computation
(3) drawing hydrograph
(4) estimation of ϕ -index
35. Components of spillway are
(1) entrance structure
(2) conduit
(3) outlet structure
(4) all of the above
36. A gravity dam can fail due to
(1) Overturning of the dam
(2) Sliding of the dam
(3) Development of tension in the dam
(4) All of the above
37. Hydrograph is a plot of
(1) rainfall intensity against time
(2) discharge against time
(3) cumulative time against rainfall
(4) cumulative discharge against time

38. According to Dicken's formula for estimating floods, the peak discharge is proportional to
- $A^{3/4}$
 - $A^{2/3}$
 - $A^{1/2}$
 - A
39. In a constant level pumping test, the rate of yield or discharge is given by
- $Q = K A i$
 - $Q = K A H / 2$
 - Both (1) and (2)
 - None of the above
40. Find the delta for a crop when its duty is 864 hectares / cumec on the field. The base period of the crop is 120 days.
- 120 cm
 - 12 cm
 - 1200 cm
 - 1.2 cm
41. If the area of the rice irrigation is equal to 800 hectares and the corresponding outlet discharge factor is equal to 1200 hectares cumec, then what is the discharge required for rice?
- 0.66 cumec sec
 - 0.066 cumec/sec
 - 6.6 cumec/sec
 - 6.06 cumec sec
42. An example of kharif crop is
- Cotton
 - Wheat
 - Gram
 - Linseed
43. Lift irrigation from surface water is called as
- Tubewell irrigation
 - Sub-surface irrigation
 - Well irrigation
 - Sprinkler irrigation
44. Palco is the
- First watering before a crop is sown
 - First watering after a crop is sown
 - Last watering
 - None of the above
45. The commonly adopted method amongst various surface irrigation methods is
- Wild flooding
 - Check flooding
 - Controlled flooding
 - Sprinkler irrigation
46. Kennedy's mean velocity is given by
- $V_{\theta} = 0.55 m D^{0.64}$
 - $V_{\theta} = 0.64 m D^{0.55}$
 - $V_{\theta} = 0.84 m D^{0.55}$
 - $V_{\theta} = 0.84 m D^{0.64}$
47. Headworks on a river causes
- aggradation on upstream
 - degradation in downstream
 - feed offtaking canal
 - All of the above
48. If the cross-sectional areas of an embankment at 30m intervals are 20, 40, 60, 50 and 30 m^2 respectively, then the volume of embankment on the basis of the prismoidal rule is

- (1) 5300 m
- (2) 8300 m
- (3) 9300 m
- (4) 9400 m

49. When the measured length is less than the actual length, the error is known as
- (1) Positive Error
 - (2) Compensating Error
 - (3) Instrumental Error
 - (4) Negative Error
50. Double plane method of trigonometric levelling is suitable when
- (1) Base of the object is accessible, object and instrument are in same plane
 - (2) Base of the object is inaccessible, object and instrument are not in same plane
 - (3) Both (1) and (2)
 - (4) None of the above
51. Tacheometry surveying is suitable for
- (1) Rough terrain
 - (2) Quick surveying with less accuracy
 - (3) Both (1) and (2)
 - (4) None of the above
52. Objective of providing a transition curve is
- (1) Gradual transition from tangent to circular curve
 - (2) Gradual introduction of super elevation
 - (3) To reduce the chance of overturning of the vehicle at the intersection of the tangent and curve
 - (4) All of the above
53. In a closed traverse, "latitudes" and "departures" of the sides were calculated, and it was observed that E latitude -1.39 and E departure = -2.17. Calculate the length and bearing of the closing error.
- (1) 2.57 m and $57^\circ 21'$
 - (2) -2.57 m and 58°
 - (3) 5.57 m and $57^\circ 21'$
 - (4) -5.57 m and $57^\circ 21'$
54. When it is not possible to set up the level midway between two points, then the difference in Elevation between them is measured by
- (1) Fly levelling
 - (2) Differential levelling
 - (3) Reciprocal levelling
 - (4) Precise levelling
55. If long chord and tangent length of a circular curve of radius 'R' are equal, then angle of deflection will be
- (1) 60°
 - (2) 120°
 - (3) 180°
 - (3) 90°
56. The eigen values of the matrix $\begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \end{bmatrix}$ are
- (1) 0,0,0
 - (2) 0,1,0
 - (3) 2, 1, 3
 - (4) -2,-1,-3
57. If $F = (x+3y) \mathbf{i} + (z-3y) \mathbf{j} + (x+2z) \mathbf{k}$ then $\nabla \cdot F$ is

- (1) 2
- (2) 1
- (3) 0
- (4) -3

58. The pair of ordinary differential equations obtained from the partial differential equation

$$\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y} \text{ by the method of variable}$$

separable is

- (1) $\frac{dX}{dx} - kX = 0, \frac{dY}{dy} + Y = 0$
- (2) $\frac{dX}{dx} + \frac{k}{X} = 0, \frac{dY}{dy} + \frac{k}{Y} = 0$
- (3) $\frac{dX}{dx} - kX = 0, \frac{dY}{dy} - \frac{k}{4}Y = 0$
- (4) $\frac{dX}{dx} + kX = 0, \frac{dY}{dy} - kY = 0$

59. The Fourier coefficient a_0 in the Fourier series expansion of $f(x) = e^x$ over the interval $(0, 2\pi)$ is

- (1) $(1 - e^{-2\pi})/\pi$
- (2) $(1 + e^{2\pi})/\pi$
- (3) $(1 + e^{-2\pi})/\pi$
- (4) $(1 - e^{2\pi})/\pi$

60. Solution of $p = \sin(y - xp)$ is

- (1) $y = c/x + \sin^{-1} c$
- (2) $y = cx + \sin^{-1} c$
- (3) $y = cx + \text{sinc}$
- (4) $y = x + \sin^{-1} e$

61. $L[H(t-3)]$, where $H(t-3)$ is a unit step function, is

- (1) $e^{-3s/s}$
- (2) e^{-3s/s^2}
- (3) $e^{3s/s}$
- (4) e^{3s/s^2}

62. The first approximate real root of the equation $x^4 - x = 10$ near $x = 2$ by Newton-Raphson method is

- (1) 3.826
- (2) 3.001
- (3) 2.912
- (4) 1.871

63. For a Poisson variates: $x : P(x=1) = P(x=2)$ then the mean of x is

- (1) 2
- (2) $\frac{1}{2}$
- (3) $\frac{1}{2}$
- (4) 2

64. Correct phonetic transcription of the word "JET" is

- (1) /jet/
- (2) /jeit/
- (3) /jt/
- (4) /d zet/

65. Choose the exact meaning of the underlined words: She exhibited remarkable sang freid during the crisis.

- (1) Temper
- (2) Irritation
- (3) Composure
- (4) Anger

66. Identify the grammatical error in the given sentence

No sooner did / the train arrived at the station/than the passengers / rushed towards it.

- (1) a
- (2) b
- (3) c
- (4) d

67. Which of the following words does not have silent letter/letters?
 (1) Mnemonic
 (2) Wreath
 (3) Brush
 (4) Hymn
68. The meanings of the Homonym 'Bark' is
 (1) leaves and sound a dog makes
 (2) a tree's outer layer and sound a dog makes
 (3) flower and sound a dog makes
 (4) roots and sound a dog makes
69. Choose the statement where underlined word is used correctly
 (1) The minister insured the victims that everything would be all right.
 (2) He ensured that the company will not have to bear any loss.
 (3) The actor got himself ensured against any accident.
 (4) The teacher insured students of good results.
70. Which of the following operator takes only integer operands?
 (1) +
 (2) *
 (3) /
 (4) %
71. What is the size of the C structure?
 (1) Size of C structure is the total bytes of all elements of structure.
 (2) C structure is always 128 bytes.
 (3) Size of C structure is the size of the largest element.
 (4) C structure is always 1024 bytes.
72. What will be the output of the following program?

```
#include <stdio.h>
void main()
{
    int x=5.3% 2;
    printf("value of x is %d",x);
}
```

 (1) value of x is 2.3
 (2) value of x is 1
 (3) value of x is 0.3
 (4) compile time error
73. The function scanf() returns
 (1) 0
 (2) 1
 (3) The number of successful read input values
 (4) The actual values read for each argument
74. The continue statement cannot be used with
 (1) switch
 (2) for
 (3) while
 (4) do while
75. All keywords in C Programming are in _____
 (1) lower case letters
 (2) upper case letters
 (3) camel case letters
 (4) all of the above
76. The maximum shear stress on the circumference of a cylindrical shell of internal radius r and thickness t when subjected to internal pressure p is

- (1) $pr/4t$
- (2) $pr/2t$
- (3) $pr/3t$
- (4) $2pr/t$

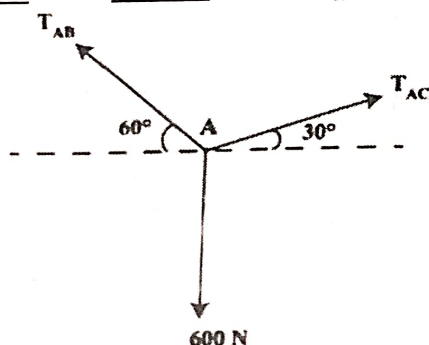
77. If P & Q are two concurrent forces having an angle θ in between them, then R the resultant will be calculated by the formula

- (1) $R = \sqrt{P^2 + Q^2}$
- (2) $R = \sqrt{P^2 + Q^2 + PQ \cos \theta}$
- (3) $R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$
- (4) $R = \sqrt{P^2 + Q^2 + 2PQ \sin \theta}$

78. The maximum strength of stones is obtained by placing the applied load

- (1) Perpendicular to the direction of the natural bed of stone
- (2) Along the direction of the natural load of stone
- (3) Both (1) and (2)
- (4) Neither (1) nor (2)

79. If point A is in equilibrium under the action of the applied forces, the values of tension T_{AB} and T_{AC} are _____ and _____ AB respectively.



- (1) 520 N and 300 N
- (2) 300 N and 520 N

- (3) 450 N and 150 N
- (4) 150 N and 450 N

80. The quickest ingredient to react when water is added to the cement is

- (1) Tricalcium aluminate ($3\text{CaO} \cdot \text{Al}_2\text{O}_3$)
- (2) Tetra calcium aluminoferrite ($4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$)
- (3) Tricalcium silicate ($3\text{CaO} \cdot \text{SiO}_2$)
- (4) Dicalcium silicate ($2\text{CaO} \cdot \text{SiO}_2$)

81. A Mohr's circle reduce to a point when the body is subjected to

- (1) Pure shear
- (2) Uniaxial stress only
- (3) Equal and opposite axial stress on two mutually $\perp^{\text{lar}}$ planes
- (4) Equal axial stresses on two mutually perpendicular planes, the plane is being free of shear

82. The safe stress for a hollow steel column which carries an axial load of 2.1×10^3 KN is 125 MN/m^2 . If the external diameter of the column is 30 cm, then the internal diameter is

- (1) 26.19 cm
- (2) 24.3 cm
- (3) 27.01 cm
- (4) 26.16 cm

83. The relation between maximum stress (σ) offered by a section, moment of resistance (M) and section modulus (Z) is given by

- (1) $M = \sigma/Z$
- (2) $M = Z/\sigma$
- (3) $M = \sigma \times Z$
- (4) $M = 1/\sigma \times Z$

84. Maximum differential settlement, in case of foundation on sandy soils is usually limited to
- (1) 10 mm
 - (2) 25 mm
 - (3) 40 mm
 - (4) 100 mm
85. Foundations in black cotton soils
- (1) are taken to depths upto which cracks do not extend
 - (2) loads are limited to 5 t/m^2
 - (3) trenches are dug on either side of foundation & filled with sand to prevent immediate contact of soil
 - (4) all of the above
86. Minimum load which will cause failure of a foundation is called
- (1) nominal strength of soil
 - (2) ultimate compressive strength of soil
 - (3) ultimate bearing capacity of soil
 - (4) all of the above
87. In pile foundation, when the load is supported by friction between pile and surrounding soil and also by resting the pile on hard stratum, such a pile is called
- (1) friction pile
 - (2) bearing pile
 - (3) friction & bearing pile
 - (4) sheet pile
88. For large spans, the most suitable type of roof preferred is
- (1) Flat roof
 - (2) Sloped roof
 - (3) Domes
 - (4) Shell roof
89. The diameter of longitudinal bars of a column should never be less than
- (1) 6 mm
 - (2) 8 mm
 - (3) 12 mm
 - (4) 10 mm
90. If the slump of the concrete is 60 mm, its workability is considered to be
- (1) medium
 - (2) low
 - (3) high
 - (4) very high
91. The percentage of lime (CaO) in ordinary portland cement will be
- (1) 60% to 67%
 - (2) 3% to 8%
 - (3) 1% to 3%
 - (4) 17% to 25%
92. A fluid which is incompressible and is having no viscosity is known as
- (1) Real fluid
 - (2) Ideal fluid
 - (3) Plastic fluid
 - (4) None of the above

93. Specific volume is the reciprocal of

- (1) Mass density
- (2) Specific density
- (3) Weight density
- (4) None of the above

94. The head of water over the centre of orifice of diameter 20 mm is 1.0 m. The actual discharge through the orifice is 0.85 litre/sec. The coefficient of discharge is

- (1) 0.81
- (2) 0.41
- (3) 0.61
- (4) 0.45

95. If the scale ratio for the linear dimensions of the model and its prototype is same, then the model is called

- (1) Distorted model
- (2) Scale ratio
- (3) Rayleigh's method
- (4) Undistorted model

96. The pressure intensity at a point in a fluid is 3.924 N/cm^2 . The corresponding height of fluid when the fluid is in water is

- (1) 1 m of water
- (2) 4 m of water
- (3) 6 m of water
- (4) 0.5 m of water

97. Find the length of the notch when $C_d = 0.62$, the head of water over a rectangular notch is 900 mm and discharge is 300 litres/sec.

- (1) 185 mm
- (2) 160 mm

(3) 141 mm

(4) 192 mm

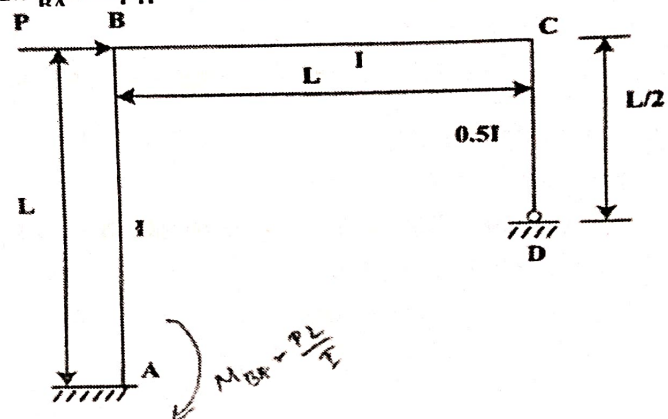
98. A horizontal venturimeter with inlet and throat diameters 30 cm and 15 cm respectively is used to measure the flow of water. The reading of differential manometer connected to the inlet and the throat is 20 cm of mercury. Determine the rate of flow. Take $C_d = 0.98$.

- (1) 125.756 lit/s
- (2) 110.001 lit/s
- (3) 98.415 lit/s
- (4) 74.213 lit/s

99. If Froude number is less than 1.0, then the flow is said to be

- (1) Sub-critical flow
- (2) Super-critical flow
- (3) Rapid flow
- (4) All of the above

100. The given figure shows a portal frame with one end fixed & other hinged. The ratio of the fixed end moments M_{RA}/M_{CD} due to side sway will be



- (1) 1.0
- (2) 2.0
- (3) 2.5
- (4) 3.0

KEY ANSWERS

1 - 3	2 - 3	3 - 1	4 - 1	5 - 2
6 - 3	7 - 1	8 - 1	9 - 3	10 - 3
11 - 4	12 - 2	13 - 1	14 - 2	15 - 4
16 - 2	17 - 1	18 - 4	19 - 1	20 - 1
21 - 1	22 - 2	23 - 3	24 - 2	25 - 2
26 - 2	27 - 1	28 - 4	29 - 3	30 - 2
31 - 4	32 - 4	33 - 4	34 - 1	35 - 4
36 - 4	37 - 2	38 - 1	39 - 3	40 - 1
41 - 1	42 - 1	43 - 3	44 - 1	45 - 3
46 - 1	47 - 4	48 - 1	49 - 4	50 - 2
51 - 3	52 - 4	53 - 1	54 - 3	55 - 4
56 - 3	57 - 3	58 - 3	59 - 1	60 - 2
61 - 1	62 - 4	63 - 1	64 - 4	65 - 3
66 - 2	67 - 3	68 - 2	69 - 2	70 - 4
71 - 1	72 - 4	73 - 3	74 - 1	75 - 1
76 - 3	77 - 3	78 - 1	79 - 1	80 - 1
81 - 4	82 - 1	83 - 3	84 - 2	85 - 4
86 - 3	87 - 3	88 - 4	89 - 3	90 - 1
91 - 1	92 - 2	93 - 1	94 - 3	95 - 4
96 - 2	97 - 4	98 - 1	99 - 1	100 - 1